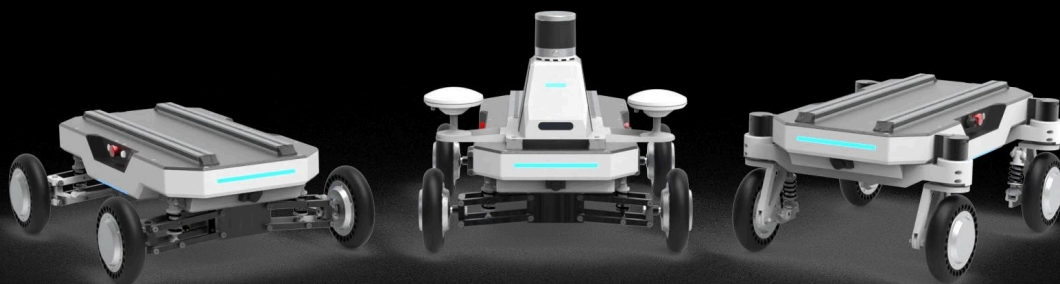


● Four-wheel-steer robot chassis

LUCHS A

USER MANUAL

Four-wheel-steer robot chassis · Configurations A-DF, A-FA, A-DA and A-4WS



VERSION 1.4

October 2026 · English edition

CHASSIS BY RJ TECH LIMITED

Autonomy by TerraAgent Robotics

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5. **Normal wear and ageing.** Loss of performance caused by normal wear and ageing under normal operating conditions.

Revision history

Version	Date	Changes
1.4	2026-10-05	New 5.5 Autonomy by TAR (option): building blocks running on real vehicles and real data.
1.3	2026-10-05	Chapters aligned with the Luchs B manual: new 3.1 Daily check, 3.3 Power off, 4.4 Control modes, 5.1 Overview, 6.1 Routine care and 6.2 Warranty; lights and payload interfaces renumbered 4.6 and 4.7; serial protocols 5.3. Section 5.4: ROS software 1.1 (ROS 2 and ROS 1, one code base with Luchs B).
1.1-1.2	2026-10-05	RJ Tech house style; ROS 2 software luchs_ros2.1.0 replaces the earlier ROS 2 driver.
1.0	October 2026	First English edition for the Luchs A product line.

About this manual

Thank you for choosing the Luchs A, an open, high-performance mobile robot chassis for research, development, prototyping and pilot projects.

This manual gives you complete and accurate information about the product and how to operate it. To use the chassis safely and get the best from it, please read this manual before you start, and keep it for future reference.

Who this manual is for

This manual covers the hardware architecture, low-level communication protocols, motion control and sensor integration. It is written mainly for:

- **Robotics and software engineers** who develop higher-level algorithms, integrate systems or build product prototypes on the Luchs A, and need a detailed understanding of its interfaces, communication protocols (such as CAN) and control commands.
- **Researchers and students** in robotics, autonomous driving and artificial intelligence who need a complete hardware and software reference platform for teaching and research in motion control, SLAM (simultaneous localisation and mapping) and multi-robot coordination.
- **Advanced makers and robotics enthusiasts** with solid electronics and programming experience who want to use the Luchs A for their own projects or competitions.
- **Technical support and maintenance engineers** who deploy, commission, maintain and troubleshoot the platform, and rely on the hardware description, self-test sequence and fault codes in this manual.

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2.1	Introduction · 2.2 Key features · 2.3 Specifications · 2.4 Chassis components · 2.5 Remote-control transmitter
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SAFETY



READ BEFORE USE

Read and understand all safety information in this manual before operating the chassis. Improper use can cause serious injury or damage to equipment.

Prohibited

Environment

- Do not operate or charge the chassis in rain or snow, in wet conditions, or anywhere it may come into contact with liquids. This can cause short circuits, electric shock or damage.
- Do not operate the chassis in the presence of flammable, explosive or highly corrosive gases, or in very dusty environments.
- Do not operate the chassis near sources of strong electromagnetic interference, such as substations or broadcast antennas. Signal loss can cause loss of control.
- Do not operate or store the chassis outside its operating temperature range of -10°C to $+45^{\circ}\text{C}$.

Operation

- Do not allow children, untrained persons, or anyone who is tired or under the influence of alcohol or medication to operate the chassis.
- Do not drive at high speed, or leave the chassis running unattended, near people or obstacles. Always keep the chassis in sight.
- Do not disassemble or service the chassis, or touch moving or hot parts (such as motors and power electronics), while it is switched on.
- Do not overload the chassis, exceed its rated speed, or use it for purposes it was not designed for.

Charging

- Use only the original charger or a charger approved by the manufacturer. An unsuitable power supply can cause fire or explosion.
- Do not leave the chassis charging unattended for long periods. If the battery swells, changes colour, overheats or smells unusual, stop charging immediately.

Mandatory

Before each use

- Check that both emergency stop buttons work, that the remote-control transmitter has enough battery, and that the chassis is undamaged.
- Operate only in an open, level area that is free of obstacles.

Charging and maintenance

- Charge with the original charger in a dry, well-ventilated place, away from flammable materials.

- Use only original spare parts. If the chassis will not be used for a long time, charge it regularly and keep the battery at 50–70 % to protect its health.
- If any fault occurs, switch the chassis off immediately and contact technical support. Do not attempt to dismantle or repair it yourself.

In an emergency

If the chassis goes out of control, catches fire or another emergency occurs:

- 1** Press an emergency stop button immediately.
- 2** Disconnect the main power.
- 3** Follow your emergency plan, move people to safety and deal with the situation.



PRODUCT OVERVIEW

2.1 Introduction

The Luchs A is an open mobile robot chassis for research, development, product prototyping and pilot deployments. It is modular, versatile and highly customisable, and supports everything from algorithm validation to full system integration.

Support is available at three levels:

- **Algorithms.** The optional navigation kit comes with ROS (Robot Operating System) pre-installed, mainstream SLAM algorithms such as GMapping and Cartographer, and the ROS navigation stack. It supports live mapping, global and local path planning, dynamic obstacle avoidance and trajectory tracking.
- **Applications.** Optional interactive software supports online and offline mapping, parameter tuning, data logging and visualisation, making it easy to evaluate algorithms and validate functions.
- **Hardware.** The chassis can carry a single-line LiDAR, depth cameras, an IMU (inertial measurement unit) and wheel encoders, and offers a wide range of expansion interfaces for further development and third-party devices.

Compact and complete, the Luchs A is suited to universities, research institutes and companies working on unmanned systems, machine vision and motion control, from teaching and research to early product development.

2.2 Key features

- **Modular, flexible design.** Standard mounting interfaces on the body make it quick to add robot arms, lifting modules and other attachments. A dedicated interface for the navigation kit provides a smooth path from basic teaching to advanced research.
- **Comprehensive, layered safety.** Ultrasonic obstacle detection, two emergency stops, IP33 protection and a self-test and warning system, plus charging supervision and over-current, over-temperature and short-circuit protection for connected devices.
- **Open hardware and software.** Industrial communication interfaces (CAN, RS-232 and RS-485) and high-power 48 V and 12 V outputs give a solid basis for integrating your own devices.

2.3 Specifications

Table 2-1 · Technical data

	A-DF Differential 4WD	A-FA Front-Ackermann 4WD	A-DA Dual-Ackermann 4WD	A-4WS Four-wheel steering
MOTION				
Special manoeuvres	Pivot turn, tight U-turn	Pivot turn, tight U-turn	Pivot turn, tight U-turn	Pivot turn, omnidirectional travel
Motion models	Skid-steer	Skid-steer, Ackermann	Skid-steer, Ackermann, dual Ackermann	Skid-steer, Ackermann, dual Ackermann, 4WS
Speed	0–11.6 km/h			

	A-DF Differential 4WD	A-FA Front-Ackermann 4WD	A-DA Dual-Ackermann 4WD	A-4WS Four-wheel steering
Minimum turning radius	0 m (pivot)	0 m (pivot) · 0.89 m (Ackermann)	0 m (pivot) · 0.445 m (dual Ackermann)	0 m
Pivot-turn clearance circle	1.01 m	1.01 m	1.01 m	0.857 m
MECHANICAL				
Dimensions (L×W×H)	800×665×292 mm	800×665×292 mm	800×665×292 mm	800×580×372 mm
Wheelbase	594 mm			
Track width	596 mm	596 mm	596 mm	500 mm
Wheel diameter	206 mm			
Ground clearance	77 mm	77 mm	77 mm	220 mm
Suspension	Double wishbone, front and rear	Double wishbone, front and rear	Double wishbone, front and rear	Independent swing arm
Chassis weight	40 kg	45 kg	50 kg	55 kg
Frame material	Carbon steel and aluminium alloy			
Mounting rails	2 × 2040 aluminium extrusion, 700 mm long, 200 mm centres			
PERFORMANCE				
Maximum payload	80 kg			
Maximum gradient	15°			
Parking brake	Electric motor hold			
Runtime	Up to 6 h unloaded · 3–5 h loaded			
Operating temperature	–10 °C to +45 °C			
Ingress protection	IP33 (standard build) · IP54–sealed A-4WS build available			
DRIVE AND POWER				
Drive motors	4 × 250 W servo hub motors			
Steering motors	None	1 × 140 W servo	1 × 140 W servo	4 × 140 W servo
Battery	LiFePO ₄ , 48 V, 15 Ah (720 Wh)			
Charging	48 V / 10 A charger included · 0–100 % in about 2 h			
Power outputs	48 V / 10 A × 1 · 12 V / 8 A × 1			
CONTROL AND COMMUNICATION				
Operating modes	Remote control · Automatic · Maintenance			
Remote control	10–channel 2.4 GHz transmitter with receiver			
Communication	CAN × 1 · RS-485 × 1 · RS-232 × 1			
Software	ROS 2 Humble / Jazzy and ROS 1 Noetic (luchs_ros 1.1: SDK, driver, robot model, simulator; one code base with Luchs B) · CAN and serial protocol documentation · DBC files			
Lighting	Status lights, breathing light, warning lights, charge-progress bar			
SAFETY				

	A-DF Differential 4WD	A-FA Front-Ackermann 4WD	A-DA Dual-Ackermann 4WD	A-4WS Four-wheel steering
Emergency stop	Two buttons, left and right			
Ultrasonic sensors	2 (front, rear)	2 (front, rear)	2 (front, rear)	6 (1 front, 1 rear, 2 per side)
Protection	Power-on self-test and fault warning · BMS · charging over-current · battery surge · actuator over-current			

Typical values. Speed, runtime and gradeability depend on payload, surface and driving style. The pivot-turn clearance circle is the diameter the chassis sweeps when it turns on the spot. Other configurations based on the standard chassis are available on request.

2.4 Chassis components

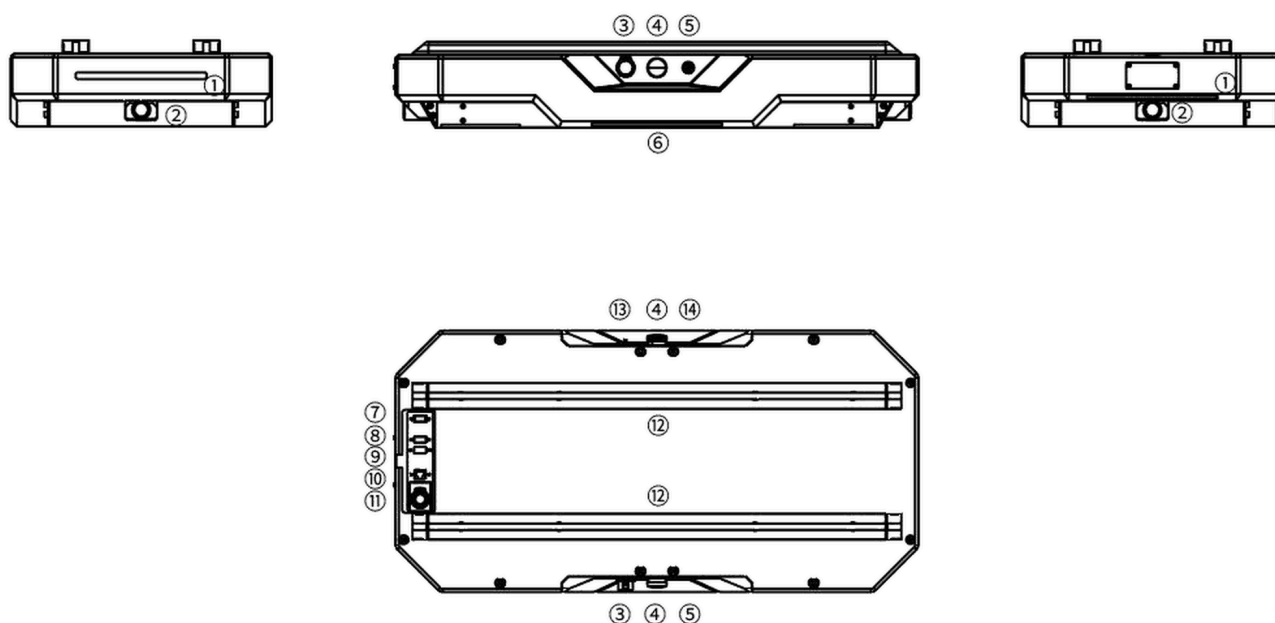


Figure 2-1 · Component locations: end, side and top views

- | | | |
|---------------------|--------------------------------------|----------------------------|
| 1 Status light | 6 Progress light | 11 Expansion power port |
| 2 Ultrasonic sensor | 7 DB9 communication port | 12 Aluminium mounting rail |
| 3 Charging port | 8 HDMI port (built-in computer) | 13 Parking brake release |
| 4 Emergency stop | 9 USB 2.0 port (built-in computer) | 14 Mode selector |
| 5 Power button | 10 Ethernet port (built-in computer) | |

The Luchs A uses a modular architecture: a common main body combined with an interchangeable drive module. The different drive configurations (differential, Ackermann and four-wheel steering) are created by fitting different, interface-compatible drive modules, so the main body and its functions are identical across the range.

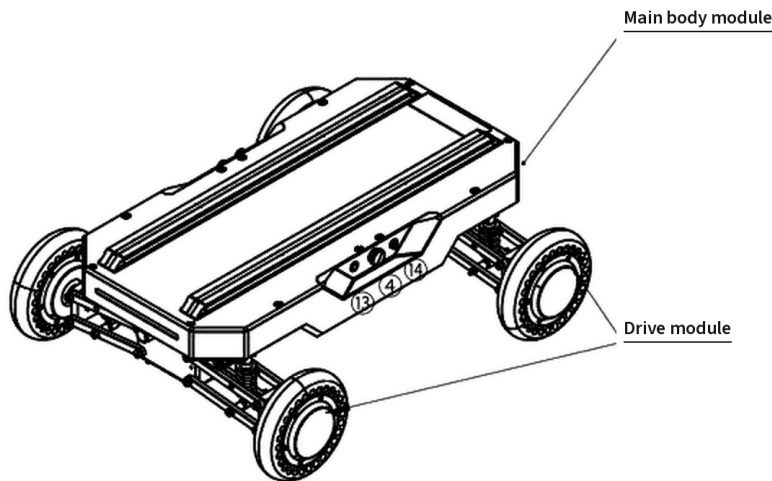


Figure 2-2 · Main body module and drive modules

NOTE

The HDMI, USB 2.0 and Ethernet ports (8, 9, 10) are only fitted when the chassis is ordered with the built-in computer.

2.5 Remote-control transmitter

The chassis is supplied with a 10-channel 2.4 GHz remote-control transmitter and a matching receiver.

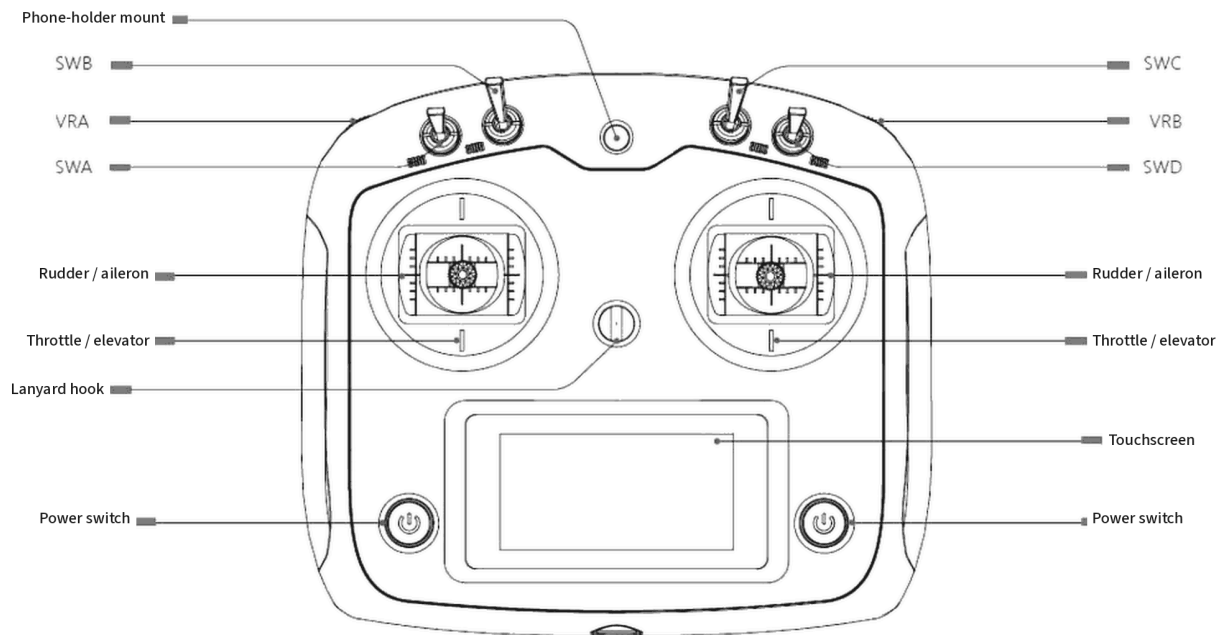


Figure 2-3 · Transmitter controls. SWA–SWD: toggle switches; VRA, VRB: thumbwheels.

Table 2-2 · Transmitter controls used with the Luchs A

Control	Function on the Luchs A	See section
Left stick	Forward and reverse; sideways travel on the A-4WS	4.3.2
Right stick	Turning (pivot or Ackermann steering, depending on configuration)	4.3.2
VRB thumbwheel	Pivot turn	4.3.2

Control	Function on the Luchs A	See section
SWA	Ultrasonic stop active / manual override	4.3.3
SWB	Speed range: 30 %, 60 % or 100 %	4.3.4
SWC	Operating mode: remote control, maintenance or automatic	4.3.5
SWD	Parking position (wheels locked against rolling)	4.3.6
Power switches	Press and hold both together to switch the transmitter on or off	4.3.1



QUICK START

Follow these steps to start your Luchs A safely. Before you begin, make sure you have read and understood Chapter 1, Safety.

3.1 Daily check

- 1 **Walk around the chassis.** Check that the area is open, level and clear, the wheels and structure are undamaged, and any load on the rails is secured.
- 2 **Check the emergency stops.** Make sure both emergency stop buttons work and are released.
- 3 **Check the transmitter.** Make sure it has enough battery and that all four toggle switches (SWA–SWD) are up.

3.2 Power on and drive by remote control

- 1 **Position the chassis.** Place it on a level, open indoor floor with no obstacles nearby.
- 2 **Release the emergency stops.** Make sure neither emergency stop button is pressed; otherwise the chassis will not start.
- 3 **Switch on.** Press the power button on the right-hand side of the chassis.
- 4 **Switch on the transmitter.** Move all four toggle switches (SWA–SWD) to the up position, then press and hold both power switches for about 2 seconds until the screen lights up.
- 5 **Select remote-control mode.** Turn the mode selector on the left-hand side of the chassis anticlockwise to the remote-control position. The status lights turn green.
- 6 **Drive.** Use the two sticks on the transmitter to move the chassis. See section 4.3 for details.

3.3 Power off

- 1 **Stop the chassis** and release the sticks. The drive motors hold the chassis in place.
- 2 **Switch off the chassis** by pressing the power button so that it is released. The status lights go dark.
- 3 **Switch off the transmitter** by pressing and holding both power switches until the screen goes dark.

3.4 Charging

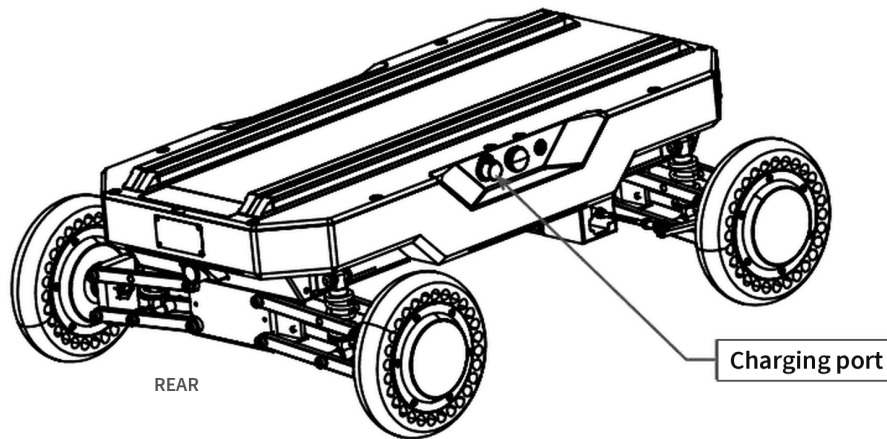


Figure 3-1 · Charging port

- 1 **Prepare.** Place the chassis in a dry, well-ventilated area away from flammable materials.
- 2 **Connect.** Plug the circular connector of the supplied charger firmly into the charging port on the right-hand side of the chassis, then connect the charger to mains power. The charger makes a running noise; this is normal. If the chassis is switched on, the front and rear lights turn orange and the light bar shows charging progress. If it is switched off, the lights stay off.
- 3 **Finish.** When the charger goes quiet, the battery is full. Disconnect the charger from mains power first, then unplug it from the chassis.

A full charge from 0 to 100 % takes about 2 hours with the standard 48 V / 10 A charger.

3.5 Connecting your computer

3.5.1 With the built-in computer

Requires the optional navigation kit with built-in computer.

- 1 **Connect your devices** to the expansion interfaces on the chassis, if you are using any.
- 2 **Start the system.** Place the chassis in a safe area, switch it on and wait for the self-test to finish.
- 3 **Select automatic mode** with the mode selector.
Safer for debugging: set the mode selector to remote control first, then switch to automatic mode with SWC on the transmitter. In an emergency, flip SWC back to maintenance or remote control to stop the chassis running away.
- 4 **Connect a display.** Connect a monitor to the HDMI port with a standard HDMI cable; the graphical interface appears automatically. If your configuration includes a display, use that instead.
- 5 **Develop and debug.** From the interface you can view system status, sensor data and chassis parameters in real time, write and debug scripts with the built-in tools, and run the SDK examples or call the API to test functions.

3.5.2 With an external computer

- 1 **Connect your computer and sensors.** Connect your industrial PC or edge computer, and any sensors, to the signal port (CAN, RS-232, RS-485) and the expansion power port using the supplied cables.
- 2 **Start the system.** Place the chassis in a safe area, switch it on and wait for the self-test to finish.

-
-
- 3 Implement the protocol.** Choose the protocol for your interface in Chapter 5 and start developing. The ROS software (section 5.4) is a ready-made starting point.



FUNCTIONS IN DETAIL

4.1 Power-on self-test

Every time the chassis is switched on, it runs a self-test to make sure the core hardware is working before it can be operated. While the test runs, the status lights fill up in purple. If the test fails, the lights flash red rapidly; contact technical support. The self-test checks the following, in order:

- Charging state.** The chassis checks whether the charging connector is plugged in. If it is (whether or not charging is in progress), the chassis enters a drive-inhibit state to prevent damage, and ignores the transmitter and the host computer until the connector is removed.
- Power system.** Battery: checks communication with the BMS (battery management system) and reads battery voltage, temperature and current to confirm they are within safe limits. Power board: checks that each supply rail (such as 12 V and 5 V) is stable.
- Communication links.** Checks the internal CAN, RS-485 and RS-232 networks.
- Motion control.** Sends a small test command to each motor driver and confirms through encoder feedback that the motor responds.
- Ultrasonic sensors.** Sends a test pulse to each sensor and analyses the echo to confirm it is working.

4.2 Controls on the chassis

4.2.1 Power button

The power button switches the main power of the chassis on and off. It latches in position and has a built-in indicator that is driven directly by the hardware, so it always shows the true supply state.

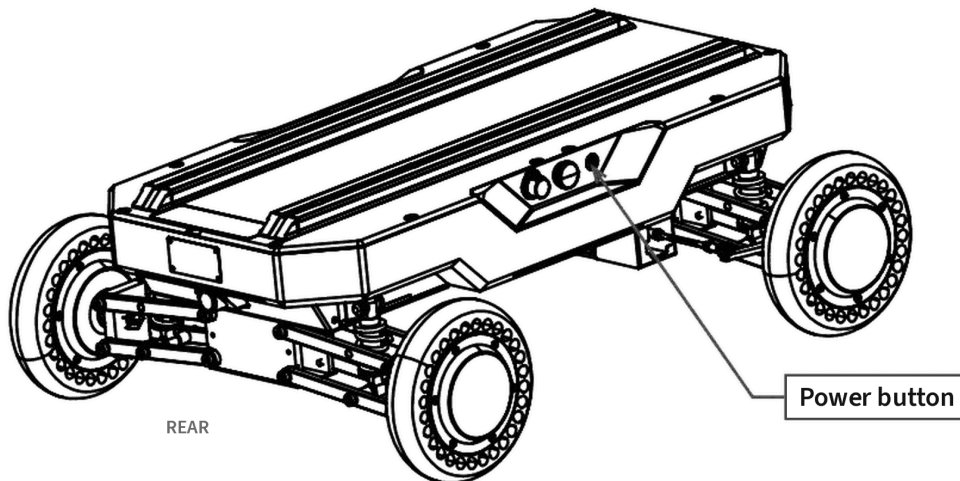


Figure 4-1 · Power button

Table 4-1 · Power state of each subsystem

State	Power button	MCU	Motion control	Peripheral circuits	Computer power	Other functions
Off	Released	Off	Off	Off	Off	Off
On	Pressed	Running	Enabled	On	On	On

4.2.2 Emergency stop buttons

The emergency stop buttons have the highest priority of all safety devices. Pressing either one cuts drive power in an emergency and brakes the chassis immediately, protecting people and equipment. There is one button on each side of the chassis.

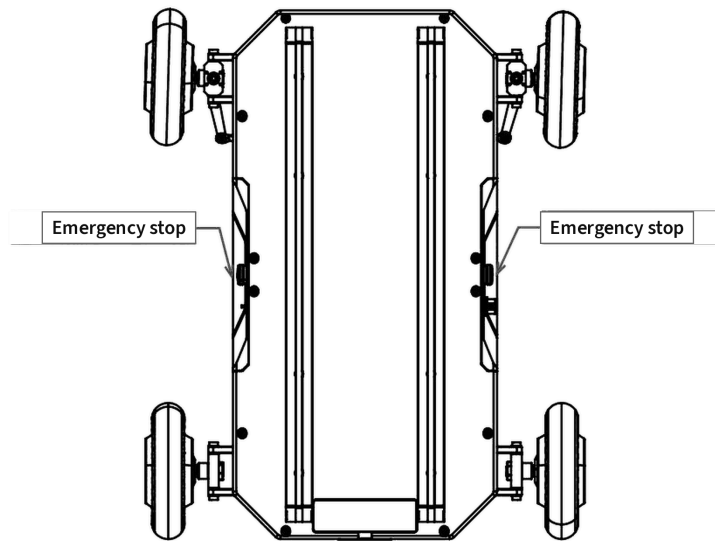


Figure 4-2 · Emergency stop buttons (top view)

Table 4-2 · Emergency stop logic

Action	Lights	Motion control	System state
Emergency stop pressed	Red, fast flashing	Motors brake; the chassis holds its position.	Enters error state
Button released by turning	Red, fast flashing	Motors brake; the chassis holds its position.	Remains in error state
Mode selector turned to maintenance	Yellow, steady	Parking brake release not pressed: motors brake (or are disabled with the holding brake applied). Parking brake release pressed: motors disabled, holding brake released.	Error and previous commands cleared; system returns to standby
Mode selector turned back to the previous mode	Remote control: green Automatic: blue	See section 4.2.3.	System resumes

4.2.3 Mode selector

The mode selector sets the basic operating mode of the chassis, so that it takes commands from the right source for the task. Turn it anticlockwise for remote control, to the centre for maintenance, and clockwise for automatic mode.

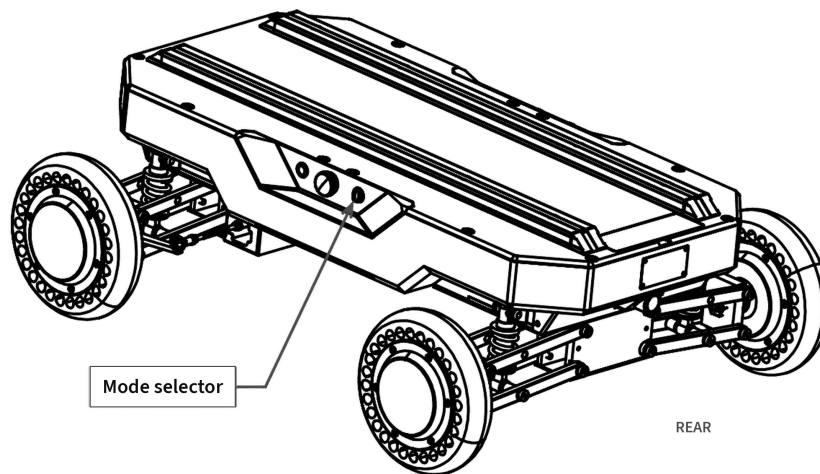


Figure 4-3 · Mode selector

Table 4-3 · Operating modes

Mode	Lights	Motion control	Parking brake release	Transmitter	Host computer	Purpose
Maintenance	Yellow	Depends on the parking brake release: pressed, motors disabled; released, motors enabled and holding position.	Active	Ignored	Conditional	Commissioning, maintenance and recovery. All error states, including recovery after an emergency stop, are cleared by switching to this mode.
Remote control	Green	Enabled	No effect	Accepted	Conditional	Driving with the transmitter. The transmitter can also switch the chassis to automatic mode (see 4.3.5).
Automatic	Blue	Enabled	No effect	Ignored	Accepted	Control by a host computer (such as an industrial PC). The chassis executes navigation and motion commands received over CAN, RS-232 or RS-485.

4.2.4 Parking brake release

The parking brake release is a latching button that directly controls braking of the drive motors. It is used in maintenance mode to release the brakes so that the chassis can be pushed by hand.

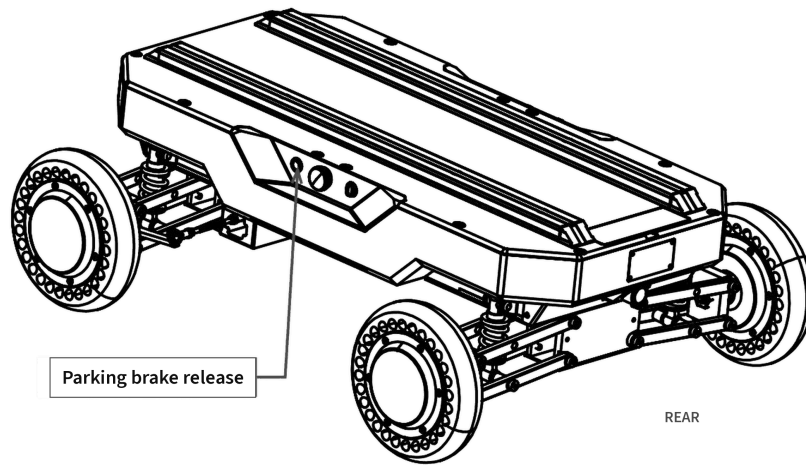


Figure 4-4 · Parking brake release

Table 4-4 · Parking brake release logic

Mode	Button released	Button pressed
Maintenance	Brakes held. Motors enabled and braking; the wheels are locked and the chassis cannot be pushed.	Brakes released. The holding brakes release and the wheels turn freely, so the chassis can be pushed by hand.
Remote control	No effect. Motors stay enabled and the chassis responds to the transmitter as normal.	
Automatic	No effect. Motors stay enabled and the chassis responds to the host computer as normal.	

4.3 Remote control

In remote-control mode (mode selector turned anticlockwise) you drive the chassis with the handheld transmitter.

4.3.1 Switching the transmitter on and off

Press and hold the two power switches at the bottom of the transmitter at the same time. The screen lights up when the transmitter is on and goes dark when it is off. The transmitter only switches on when all four toggle switches, SWA to SWD, are in the up position.

4.3.2 Driving

Drive the chassis with the two sticks and the VRB thumbwheel at the top right of the transmitter.

Table 4-5 · Stick functions

Control	Movement	A-DF	A-FA · A-DA	A-4WS
Left stick	Forward / back	Forward / reverse	Forward / reverse	Forward / reverse
Left stick	Left / right	No response	No response	Moves sideways left / right
Right stick	Left / right	Pivot turn anticlockwise / clockwise	Steers left / right (Ackermann)	Pivot turn anticlockwise / clockwise

Control	Movement	A-DF	A-FA · A-DA	A-4WS
VRB thumbwheel	Left / right	Pivot turn anticlockwise / clockwise	Pivot turn anticlockwise / clockwise	Pivot turn anticlockwise / clockwise

NOTE

In normal remote-control use the A-4WS drives in four-wheel-steering mode. As soon as it receives a command over CAN or the serial port that selects a specific motion model, remote control switches to that model as well. This makes it easier to check odometry when you are developing autonomous navigation.

4.3.3 Ultrasonic stop and manual override (SWA)

In remote-control mode the ultrasonic sensors constantly watch for obstacles. If an obstacle comes closer than 20 cm, the chassis makes a safety stop and holds its position. Once you have confirmed it is safe, you can override the stop with SWA.

Table 4-6 · SWA logic

SWA	Obstacle within safety distance	Status light on that side	Transmitter commands	Chassis
Up	No	Green	Accepted	Enabled, moves normally
Up	Yes	Red, steady	Ignored	Emergency brake, holds position
Down	No	Green	Accepted	Enabled, moves normally
Down	Yes	Red, steady	Accepted	Brake released; accepts commands and can move

4.3.4 Speed range (SWB)

SWB selects one of three speed ranges, for example a slow range for testing in tight spaces and the full range for open areas.

Table 4-7 · SWB speed ranges

SWB	Speed in remote-control mode
Up	30 % of maximum speed
Centre	60 % of maximum speed
Down	100 % of maximum speed

4.3.5 Switching modes from the transmitter (SWC)

When the mode selector is in the remote-control position, SWC lets you switch the operating mode from the transmitter, and with it the source of control, for testing, manual driving or automatic operation.

Table 4-8 · SWC logic

SWC	Mode	Host commands	Transmitter commands	Description
Up	Remote control	Ignored	Accepted	The transmitter has full control. Use for all manual driving.

SWC	Mode	Host commands	Transmitter commands	Description
Centre	Maintenance	Ignored	Only SWC	For testing and maintenance. Motion is restricted and the chassis only responds to the mode switch, for safety.
Down	Automatic	Accepted	Ignored	The host computer (such as an industrial PC) controls the chassis, for example for autonomous navigation.

4.3.6 Parking position (SWD)

SWD turns the wheels into a parking position in remote-control mode, so that a heavily loaded chassis cannot roll when it is standing on a slope.

Table 4-9 · SWD logic

SWD	A-DF	A-FA · A-DA	A-4WS
Up	No response	No response	No response
Down	No response	The Ackermann steering turns fully to one side; the chassis ignores all inputs except SWD.	The four wheels turn into an X pattern; the chassis ignores all inputs except SWD.

4.3.7 Transmitter battery

When the transmitter is on, the TX icon on its screen shows the remaining battery. The transmitter warns you when the battery is low; replace the batteries promptly.

- 1 Open the battery cover on the back of the transmitter.
- 2 Insert four fully charged AA batteries, making sure the terminals make good contact.
- 3 Close the battery cover.

4.4 Control modes

The chassis always knows who is in control, and it, not the host computer, decides which source it accepts. The mode is set with the mode selector on the chassis (section 4.2.3) or, from remote control, with SWC on the transmitter (section 4.3.5).

Table 4-10 · Control modes

Mode	How to enter	Accepts	Purpose
Remote control	Mode selector anticlockwise, or SWC up	Transmitter	Manual driving, moving the chassis, supervising tests. Status lights green.
Automatic	Mode selector clockwise, or SWC down	Host computer	Your software drives the chassis over CAN, RS-232 or RS-485. Status lights blue.
Maintenance	Mode selector in the centre, or SWC centre	Neither (SWC only)	Commissioning and recovery: errors, including an emergency stop, are cleared here; the parking brake release lets you push the chassis. Status lights yellow.

REMOTE CONTROL ALWAYS WINS

With the mode selector on remote control, flipping SWC up takes control back from the host computer immediately.

4.5 Safety functions

4.5.1 Emergency stop

The emergency stop buttons are the highest-priority safety devices and stop all operation immediately. Pressing either button:

- cuts the motion-control signals immediately,
- brakes the drive system, and
- locks the chassis in a safe state.

To recover:

- 1 Make sure the hazard has been removed.
- 2 Turn the emergency stop button clockwise to release it.
- 3 Turn the mode selector to another position, then back to the original mode.
- 4 The system clears the error automatically and resumes normal operation.

4.5.2 Hardware monitoring

The system continuously monitors its key components. If it detects a fault, it stops motion control immediately and flashes the status lights red, so that faults are caught early and isolated. It monitors:

- motor condition (overload, stall),
- validity of sensor data,
- power-module parameters, and
- communication links.

4.5.3 Battery management (BMS)

The BMS monitors charging, discharging, voltage and temperature. In the event of over-charge, over-discharge, over-current or abnormal temperature, it automatically disconnects power or adjusts the charging current, preventing damage or accidents and extending battery life.

4.5.4 Ultrasonic collision protection

The ultrasonic sensors monitor the surroundings in real time. When an obstacle is closer than 20 cm, the chassis responds according to the operating mode:

- **Remote control:** with SWA up, the chassis stops and transmitter commands are ignored; with SWA down, you can override the stop manually.
- **Automatic:** the ultrasonic distances are reported in real time over the protocol, and the chassis follows the commands from your navigation system, which fuses this data with its other sensors.

The serial protocols also offer an indoor collision-avoidance setting (30 cm stop distance) and an outdoor setting (50 cm); see Chapter 5.

4.5.5 Charging interlock

As soon as the charging connector is plugged in, whether the chassis is switched on or the charger is powered, the chassis disables the drive (emergency brake and park) and ignores all commands from the transmitter and the host computer. Only the computer power, the expansion power outputs and the charging lights stay on. This prevents the chassis from moving while connected to the charger and ensures safe charging.

4.5.6 Output protection

Every power channel of the chassis has real-time current monitoring with programmable soft cut-off. Critical outputs (drive, navigation and brakes) also have hardware protection against overload, short circuit and over-temperature. If any of these protections trips, restart the chassis.

4.6 Lights

The chassis has two groups of lights. The **status lights** at the front and rear show the core system state, such as the operating mode and alarms. The **progress lights** on both sides show processes and values, such as self-test progress and battery level.

Table 4-11 · Operating-mode lights

State	Lights	Meaning
Self-test	Purple, filling up	The hardware self-test runs at start-up. When the bar is full, the self-test is complete.
Maintenance	Yellow, steady	Testing state: all hardware is powered but motion is restricted.
Remote control	Green, steady	The chassis accepts transmitter commands only. After 2 minutes without a command the light breathes green (standby).
Automatic	Blue, steady	The chassis executes host commands. After 2 minutes without a command the light breathes blue (standby).

Table 4-12 · Alarm lights

State	Lights	Meaning and action
Ultrasonic obstacle	Status light on that side red, steady	An obstacle was detected closer than 20 cm in remote-control mode and the chassis has stopped. Clear the obstacle, or override with SWA.
Emergency stop	Red, fast flashing	An emergency stop button has been pressed and the chassis has braked. Check for hazards, then reset manually.
Safety alarm	Red, fast flashing	A serious fault has been detected (for example, loss of communication or a power fault). Stop and have the chassis serviced.

Table 4-13 · Charging and battery-level lights

State	Lights	Meaning
Charging	Status lights orange, steady; progress lights orange, running	Shown while charging with the chassis switched on.

State	Lights	Meaning
Battery level	Progress lights show segments in the colour of the current mode	75–100 %: all segments 50–75 %: three quarters 25–50 %: half 10–25 %: one quarter 0–10 %: one quarter, and the centre LEDs of the progress and status lights flash slowly red: charge now

4.7 Payload interfaces

4.7.1 Power and signal ports

The expansion bay is under the magnetic cover at the rear of the top plate (lever it open gently with a flat-blade screwdriver). It is designed for connecting external devices and for further development.

Table 4-14 · Expansion interfaces and pin assignments

Port	Connector	Qty	Purpose	Pin assignment
Signal port	DB9	1	CAN, RS-485 and RS-232 communication	Pin 1 (black): RS-232 RX Pin 2 (brown): RS-232 TX Pin 3 (red): GND Pin 4 (orange): CAN H Pin 5 (yellow): CAN L Pin 6 (green): RS-485 A Pin 7 (blue): RS-485 B Pin 8 (grey): GND Pin 9 (white): GND
Expansion power port	Circular connector, LP20, 4-pin	1	Power for external devices: 48 V / 10 A and 12 V / 8 A	Pin 1: 48 V + (to XT60 female) Pin 2: 48 V – (to XT60 female) Pin 3: 12 V + (to DC 5.5 × 2.1 mm plug) Pin 4: 12 V – (to DC 5.5 × 2.1 mm plug)
Computer port	HDMI	1	Display output for the computer interface	Standard HDMI
Computer port	USB 2.0	1	Power and data for peripherals	Standard USB Type-A
Computer port	Ethernet	1	Network connection for the computer	Standard RJ45

The HDMI, USB 2.0 and Ethernet ports are only fitted when the chassis is ordered with the built-in computer.

4.7.2 Mounting rails

Two parallel 2040 aluminium extrusions run along the top of the chassis at 200 mm centres. Use standard 40-series slot accessories to mount your own structures or sensor brackets. Ready-made mounting accessories are available on request.

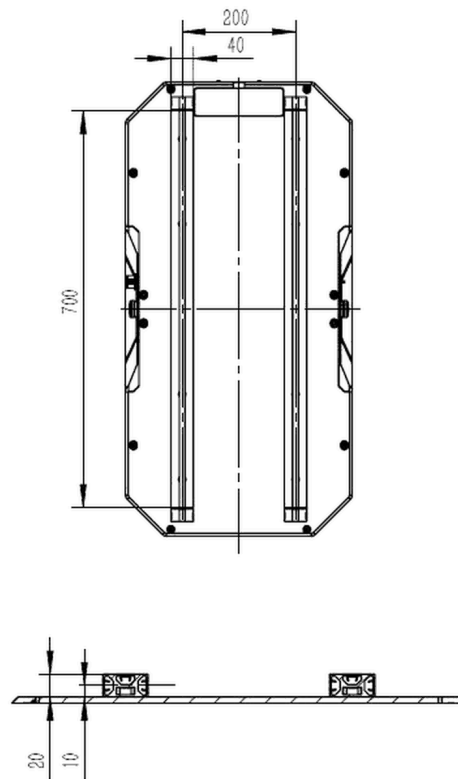


Figure 4-5 · Mounting rails: 200 mm centres, 700 mm long. Dimensions in mm.

COMMUNICATION INTERFACES AND SOFTWARE

The Luchs A provides three industrial communication interfaces, CAN, RS-485 and RS-232, so you can control the chassis and read its sensor data with your own software. The same definitions are available as spreadsheets: Luchs_A_CAN_Protocol_v1.1.xlsx and Luchs_A_Serial_Protocol_v1.0.xlsx.

VALUES UNDER CONFIRMATION

Fields shaded yellow carry values that are still being confirmed. Check them against live data from your unit before relying on them.

5.1 Overview

Table 5-1 · Interfaces for your software

Interface	Use	Documentation
CAN, 1 Mbit/s	Main interface: motion, steering, lamps, all feedback	Section 5.2 · Luchs_A_CAN_Protocol_v1.1.xlsx, Luchs_A_CAN_v1.1.dbc
RS-485 / RS-232	Alternative serial control	Section 5.3 · Luchs_A_Serial_Protocol_v1.0.xlsx
ROS 2 / ROS 1	Ready-made driver on the CAN interface	Section 5.4 · luchs_ros2_v1.1.0.zip, luchs_ros1_v1.1.0.zip

The Luchs A protocol uses the same identifiers and layouts as the core frames of the Luchs Chassis Interface (LCI) used on Luchs B, and the ROS software drives both chassis from one code base.

5.2 CAN protocol

Standard	CAN 2.0B
Bit rate	1 Mbit/s
Frame format	Standard frame, 11-bit identifier, 8-byte payload
Byte order	Big-endian: the high byte of a multi-byte signal is sent first
Physical value	Raw value × factor. Speeds and angles are signed 16-bit integers with a factor of 0.001 (m/s, rad/s, rad).
Directions	Host → Chassis: commands from your computer. Chassis → Host: feedback from the chassis controller.

Table 5-2 · CAN messages by configuration

CAN ID · Message	A-DF	A-FA	A-DA	A-4WS	Remarks
0x5A0 Motion command	●	●	●	●	vy (bytes 2–3) on A-4WS only

CAN ID · Message	A-DF	A-FA	A-DA	A-4WS	Remarks
0x5A1 Gear and Ackermann steering command	—	●	●	—	Rear angle used in dual-Ackermann mode
0x5A2 Wheel steering-angle command	—	—	—	●	
0x5A3 Auxiliary device command (lamps)	○	○	○	○	Custom builds only
0x5A4 Timestamp	opt.	opt.	opt.	opt.	Optional; needed only with the event data recorder option
0x5B5 Motion feedback	●	●	●	●	vy on A-4WS only
0x5B6 Gear and Ackermann steering feedback	—	●	●	—	
0x5B7 Wheel steering-angle feedback	—	—	—	●	
0x5B8 Protection status feedback	●	●	●	●	Standard units: front/rear ultrasonic only, no bumpers
0x5B9 Drive-motor speed feedback	—	—	—	●	Specified for A-4WS; may also be present on other variants
0x5BA Motor fault code feedback	●	●	●	●	
0x5BB Battery (BMS) feedback	●	●	●	●	
0x5BC Lamp status feedback	○	○	○	○	Custom builds only

● applies · opt. optional · ○ custom builds only, not used on the standard Luchs A · — not applicable

Table 5-3 · CAN message definitions

Byte	Signal	Type	Factor	Unit	Values	Notes
0x5A0 MOTION COMMAND · HOST → CHASSIS · ALL						
0-1	Linear velocity x (vx)	int16	0.001	m/s	±32.767	
2-3	Linear velocity y (vy)	int16	0.001	m/s	±32.767	Lateral velocity is available on A-4WS only. Other variants: send 0.
4-5	Yaw rate (wz)	int16	0.001	rad/s	±32.767	
6	Kinematic model	uint8	—	—	0x01 = Differential 0x02 = Single Ackermann (front) 0x03 = Four-wheel steering 0x04 = Dual Ackermann	Default: 0x01 (differential). A-DF: 0x01 only. A-FA: 0x01, 0x02. A-DA: 0x01, 0x02, 0x04. A-4WS: all models.
7	Control mode	uint8	—	—	0x00 = Autonomous mode	Only value defined.
0x5A1 GEAR AND ACKERMANN STEERING COMMAND · HOST → CHASSIS · A-FA, A-DA						

Byte	Signal	Type	Factor	Unit	Values	Notes
0	Gear	uint8	—	—	0 = Gear disabled 1 = P (park) 2 = R (reverse) 3 = N (neutral) 4 = D (drive)	
1–2	Front Ackermann steering angle	int16	0.001	rad	±32.767	
3–4	Rear Ackermann steering angle	int16	0.001	rad	±32.767	Rear-axle steering; applies to dual-Ackermann mode (A-DA).
5–7	Reserved	—	—	—	Write 0x00	
0X5A2 WHEEL STEERING-ANGLE COMMAND · HOST → CHASSIS · A-4WS						
0–1	Steering angle, left front (LF)	int16	0.001	rad	±32.767	Byte order in this frame: LF, LR, RF, RR. The reference ROS drivers (v1 and v2) transmit lamp commands on 0x5A2 and the wheel steering angles on 0x5A3 (order LF, RF, LR, RR), i.e. the opposite of this table. Confirm against the firmware on your unit before commanding individual wheel angles.
2–3	Steering angle, left rear (LR)	int16	0.001	rad	±32.767	
4–5	Steering angle, right front (RF)	int16	0.001	rad	±32.767	
6–7	Steering angle, right rear (RR)	int16	0.001	rad	±32.767	
0X5A3 AUXILIARY DEVICE COMMAND (LAMPS) · HOST → CHASSIS · CUSTOM BUILDS ONLY – NOT USED ON THE STANDARD LUCHS A						
0	Turn indicator	uint8	—	—	0x00 = Off 0x01 = Left 0x02 = Right	See the note on 0x5A2.
1	Reverse lamp	uint8	—	—	0x00 = Normal 0x01 = Reversing	
2	Fault lamp	uint8	—	—	0x00 = Normal 0x01 = Host system fault	
3–7	Reserved	—	—	—	Write 0x00	
0X5A4 TIMESTAMP · HOST → CHASSIS · ALL (OPTIONAL)						
0–3	Timestamp, high 32 bits	uint32	—	See notes		Only required on units fitted with the event data recorder option; otherwise it may be omitted.

Byte	Signal	Type	Factor	Unit	Values	Notes
4-7	Timestamp, low 32 bits	uint32	—	See notes		Time base and epoch are not specified. The reference ROS driver prepares milliseconds since the Unix epoch. Verify on your unit.
0X5B5 MOTION FEEDBACK · CHASSIS → HOST · ALL						
0-1	Linear velocity x (vx)	int16	0.001	m/s	±32.767	
2-3	Linear velocity y (vy)	int16	0.001	m/s	±32.767	A-4WS only.
4-5	Yaw rate (wz)	int16	0.001	rad/s	±32.767	
6	Kinematic model	uint8	—	—	0x01 = Differential 0x02 = Ackermann 0x03 = Four-wheel steering 0x04 = Dual Ackermann	
7	Control mode	uint8	—	—	11 = Manual (remote control) 22 = Autonomous	The radix of 11 22 is not stated in the CAN specification; the serial protocol uses 0x11 0x22. Verify on your unit.
0X5B6 GEAR AND ACKERMANN STEERING FEEDBACK · CHASSIS → HOST · A-FA, A-DA						
0	Gear	uint8	—	—	0 = Gear disabled 1 = P (park) 2 = R (reverse) 3 = N (neutral) 4 = D (drive)	
1-2	Front Ackermann steering angle	int16	0.001	rad	±32.767	
3-4	Rear Ackermann steering angle	int16	0.001	rad	±32.767	
5-7	Reserved	—	—	—	Write 0x00	
0X5B7 WHEEL STEERING-ANGLE FEEDBACK · CHASSIS → HOST · A-4WS						
0-1	Steering angle, left front (LF)	int16	0.001	rad	±32.767	Byte order in this frame: LF, LR, RF, RR.
2-3	Steering angle, left rear (LR)	int16	0.001	rad	±32.767	
4-5	Steering angle, right front (RF)	int16	0.001	rad	±32.767	
6-7	Steering angle, right rear (RR)	int16	0.001	rad	±32.767	
0X5B8 PROTECTION STATUS FEEDBACK · CHASSIS → HOST · ALL						

Byte	Signal	Type	Factor	Unit	Values	Notes
0	Ultrasonic stop triggered	uint8	—	—	0x00 = Not triggered 0x01 = Left 0x02 = Right 0x03 = Front 0x04 = Rear	
1	Bumper triggered	uint8	—	—	0x00 = Not triggered 0x01 = Front 0x02 = Rear	The standard Luchs A has no bumpers (contact edges); reads 0x00.
2	Emergency stop triggered	uint8	—	—	0x00 = Not triggered 0x01 = Left button 0x02 = Right button	
3	Chassis fault code	uint8	—	—	See sheet "Error Codes"	
4	Ultrasonic distance, front	uint8	0.01	cm	0-255 raw	Documented as raw 100 in cm, which limits the range to 2.55 cm and cannot represent the stop distance. Verify on your unit — scaling under confirmation.
5	Ultrasonic distance, rear	uint8	0.01	cm	0-255 raw	See byte 4. Verify on your unit — scaling under confirmation.
6	Ultrasonic distance, left	uint8	0.01	cm	0-255 raw	The standard Luchs A has front and rear ultrasonic sensors only. Verify on your unit — scaling under confirmation.
7	Ultrasonic distance, right	uint8	0.01	cm	0-255 raw	The standard Luchs A has front and rear ultrasonic sensors only. Verify on your unit — scaling under confirmation.
0X5B9 DRIVE-MOTOR SPEED FEEDBACK · CHASSIS → HOST · A-4WS						
0-1	Motor speed, left front (LF)	int16	0.001	rad/s	±32.767	Documented byte order: LF, RF, LR, RR. The reference ROS driver decodes LF, LR, RF, RR. Order and scaling: Verify on your unit — scaling under confirmation.
2-3	Motor speed, right front (RF)	int16	0.001	rad/s	±32.767	See bytes 0-1. Verify on your unit — scaling under confirmation.
4-5	Motor speed, left rear (LR)	int16	0.001	rad/s	±32.767	See bytes 0-1. Verify on your unit — scaling under confirmation.
6-7	Motor speed, right rear (RR)	int16	0.001	rad/s	±32.767	See bytes 0-1. Verify on your unit — scaling under confirmation.
0X5BA MOTOR FAULT CODE FEEDBACK · CHASSIS → HOST · ALL						

Byte	Signal	Type	Factor	Unit	Values	Notes
0-3	Motor fault code, high 32 bits	uint32	—	—	Bit field	Bit assignments are not yet published; contact RJ Tech Limited support.
4-7	Motor fault code, low 32 bits	uint32	—	—	Bit field	See bytes 0-3.
0X5BB BATTERY (BMS) FEEDBACK · CHASSIS → HOST · ALL						
0-1	Battery state of charge	int16	0.001	%	0-100 %	Documented as raw 1000; an int16 at this scale cannot exceed 32.767 %. Verify on your unit — scaling under confirmation.
2-3	Battery temperature	int16	0.001	°C		Documented as raw 1000; an int16 at this scale cannot exceed 32.767 °C. Verify on your unit — scaling under confirmation.
4-5	Charging progress	uint16	1	%	0-100 %	Unscaled value.
6-7	Battery voltage	uint16	0.1	V	0-6553.5 V	Scaling confirmed by measurement with the reference ROS 2 driver (raw 105 = 10.5 V).
0X5BC LAMP STATUS FEEDBACK · CHASSIS → HOST · CUSTOM BUILDS ONLY – NOT USED ON THE STANDARD LUCHS A						
0	Turn indicator	uint8	—	—	0x00 = Off 0x01 = Left 0x02 = Right	
1	Reverse lamp	uint8	—	—	0x00 = Normal 0x01 = Reversing	
2	Fault lamp	uint8	—	—	0x00 = Normal 0x01 = Host system fault	
3	Light / rain sensor	uint8	—	—	0x00 = Dark, dry 0x01 = Dark, rain 0x10 = Light, dry 0x11 = Light, rain	Only on builds fitted with a light/rain sensor.
4-7	Reserved	—	—	—	Write 0x00	

5.3 Serial protocols (RS-485 / RS-232)

Interfaces	RS-485 and RS-232 on the DB9 signal port under the magnetic cover at the rear of the top plate: pin 1 = RS-232 RX, pin 2 = RS-232 TX, pin 6 = RS-485 A, pin 7 = RS-485 B, pins 3, 8, 9 = GND
Serial settings	Baud rate: see firmware release notes. Data bits, parity and stop bits: see firmware release notes.
Frame structure	Header byte + payload + tail byte, fixed length per configuration. No length field and no checksum.

Byte order	int16("short") values are transmitted high byte first.
Scaling	Speeds and angles: raw = physical value × 1000 (0.001 m/s, 0.001 rad/s, 0.001 rad per bit).
External serial control	Command byte "External serial control" selects whether the serial port is used for control: 0x01 when an external compute unit drives the chassis, 0x00 when the built-in compute unit is used.

Each configuration uses its own frame header and tail:

Configuration	Command header	Command tail	Feedback header	Feedback tail	Frame length
Luchs A-DF	0xA1	0x1A	0xA2	0x2A	13 bytes
Luchs A-FA, Luchs A-DA	0xB1	0x1B	0xB2	0x2B	17 bytes
Luchs A-4WS	0xC1	0x1C	0xC2	0x2C	21 bytes

Confirm the frame header and tail against the firmware release shipped with your unit. Some earlier documentation uses 0xA1 / 0x1A (command) and 0xA2 / 0x2A (feedback) for all configurations.

5.3.1 Luchs A-DF

Table 5-4 · Command frame (Host → Chassis) — 13 bytes

Byte	Field	Type	Scaling / unit	Values	Notes
0	Frame header	uint8	—	0xA1	Confirm against your firmware release.
1-2	Linear velocity x (vx)	int16	raw = value × 1000 (0.001 per bit) m/s	±32.767	
3-4	Linear velocity y (vy)	int16	raw = value × 1000 (0.001 per bit) m/s	±32.767	Not used on A-DF; send 0.
5-6	Yaw rate (wz)	int16	raw = value × 1000 (0.001 per bit) rad/s	±32.767	Yaw rate. Earlier documentation lists the unit as m/s; the field is an angular rate in rad/s. Verify on your unit.
7	Ultrasonic collision avoidance	uint8	—	0x00 = Disabled 0x01 = Indoor collision avoidance (30 cm stop distance) 0x02 = Outdoor collision avoidance (50 cm stop distance)	
8	Turn indicator	uint8	—	0x00 = No control 0x01 = Left 0x02 = Right 0x03 = Hazard (both)	

Byte	Field	Type	Scaling / unit	Values	Notes
9	External serial control	uint8	—	0x00 = External serial control disabled (built-in compute unit) 0x01 = External serial control enabled (external compute unit)	
10	Reserved	uint8	—	0x00	
11	Reserved	uint8	—	0x00	
12	Frame tail	uint8	—	0x1A	Confirm against your firmware release.

Table 5-5 · Feedback frame (Chassis → Host) — 13 bytes

Byte	Field	Type	Scaling / unit	Values	Notes
0	Frame header	uint8	—	0xA2	Confirm against your firmware release.
1-2	Linear velocity x (vx)	int16	raw = value × 1000 (0.001 per bit) m/s	±32.767	
3-4	Linear velocity y (vy)	int16	raw = value × 1000 (0.001 per bit) m/s	±32.767	
5-6	Yaw rate (wz)	int16	raw = value × 1000 (0.001 per bit) rad/s	±32.767	Yaw rate. Earlier documentation lists the unit as m/s; the field is an angular rate in rad/s. Verify on your unit.
7	Battery level	uint8	—	0-100 Low-battery warning below 15	Percent.
8	Control mode	uint8	—	0x11 = Remote control 0x22 = Autonomous	
9	Fault and charging status	uint8	—	0x00 = No fault 0x01 = Ultrasonic stop 0x02 = Emergency stop 0x03 = Parking brake engaged 0x04 = Charging (charger connected) 0x05-0x0E = Chassis fault (see sheet "Error Codes")	
10	Reserved	uint8	—	0x00	
11	Reserved	uint8	—	0x00	
12	Frame tail	uint8	—	0x2A	Confirm against your firmware release.

5.3.2 Luchs A-FA and A-DA

Table 5-6 · Command frame (Host → Chassis) — 17 bytes

Byte	Field	Type	Scaling / unit	Values	Notes
0	Frame header	uint8	—	0xB1	Confirm against your firmware release.
1-2	Linear velocity x (vx)	int16	raw = value × 1000 (0.001 per bit) m/s	±32.767	
3-4	Linear velocity y (vy)	int16	raw = value × 1000 (0.001 per bit) m/s	±32.767	Not used on A-FA A-DA; send 0.
5-6	Yaw rate (wz)	int16	raw = value × 1000 (0.001 per bit) rad/s	±32.767	Yaw rate. Earlier documentation lists the unit as m/s; the field is an angular rate in rad/s. Verify on your unit.
7-8	Front Ackermann steering angle	int16	raw = value × 1000 (0.001 per bit) rad	±32.767	
9-10	Rear Ackermann steering angle	int16	raw = value × 1000 (0.001 per bit) rad	±32.767	Rear-axle steering; applies to A-DA.
11	Ultrasonic collision avoidance	uint8	—	0x00 = Disabled 0x01 = Indoor collision avoidance (30 cm stop distance) 0x02 = Outdoor collision avoidance (50 cm stop distance)	
12	Turn indicator	uint8	—	0x00 = No control 0x01 = Left 0x02 = Right 0x03 = Hazard (both)	
13	Kinematic model	uint8	—	0x01 = Differential 0x02 = Ackermann	Numbering differs from the CAN protocol (CAN: 0x02 single Ackermann, 0x04 dual Ackermann).
14	External serial control	uint8	—	0x00 = External serial control disabled (built-in compute unit) 0x01 = External serial control enabled (external compute unit)	
15	Reserved	uint8	—	0x00	
16	Frame tail	uint8	—	0x1B	Confirm against your firmware release.

Table 5-7 · Feedback frame (Chassis → Host) — 17 bytes

Byte	Field	Type	Scaling / unit	Values	Notes
0	Frame header	uint8	—	0xB2	Confirm against your firmware release.
1-2	Linear velocity x (vx)	int16	raw = value × 1000 (0.001 per bit) m/s	±32.767	
3-4	Linear velocity y (vy)	int16	raw = value × 1000 (0.001 per bit) m/s	±32.767	
5-6	Yaw rate (wz)	int16	raw = value × 1000 (0.001 per bit) rad/s	±32.767	Yaw rate. Earlier documentation lists the unit as m/s; the field is an angular rate in rad/s. Verify on your unit.
7-8	Front Ackermann steering angle	int16	raw = value × 1000 (0.001 per bit) rad	±32.767	
9-10	Rear Ackermann steering angle	int16	raw = value × 1000 (0.001 per bit) rad	±32.767	
11	Battery level	uint8	—	0-100 Low-battery warning below 15	Percent.
12	Control mode	uint8	—	0x11 = Remote control 0x22 = Autonomous	
13	Fault status	uint8	—	0x00 = No fault 0x01 = Ultrasonic stop 0x02 = Emergency stop 0x03 = Parking brake engaged 0x04 = Charging (charger connected) 0x05-0x0E = Chassis fault (see sheet "Error Codes")	
14	Kinematic model	uint8	—	0x01 = Differential 0x02 = Ackermann	
15	Reserved	uint8	—	0x00	
16	Frame tail	uint8	—	0x2B	Confirm against your firmware release.

5.3.3 Luchs A-4WS

Table 5-8 · Command frame (Host → Chassis) — 21 bytes

Byte	Field	Type	Scaling / unit	Values	Notes
0	Frame header	uint8	—	0xC1	Confirm against your firmware release.

Byte	Field	Type	Scaling / unit	Values	Notes
1-2	Linear velocity x (vx) — resultant speed	int16	raw = value × 1000 (0.001 per bit) m/s	±32.767	
3-4	Linear velocity y (vy)	int16	raw = value × 1000 (0.001 per bit) m/s	±32.767	
5-6	Yaw rate (wz) — resultant yaw rate	int16	raw = value × 1000 (0.001 per bit) rad/s	±32.767	Yaw rate. Earlier documentation lists the unit as m/s; the field is an angular rate in rad/s. Verify on your unit.
7-8	Steering angle, left front (LF)	int16	raw = value × 1000 (0.001 per bit) rad	±32.767	Also used as the front Ackermann angle.
9-10	Steering angle, right front (RF)	int16	raw = value × 1000 (0.001 per bit) rad	±32.767	
11-12	Steering angle, left rear (LR)	int16	raw = value × 1000 (0.001 per bit) rad	±32.767	Also used as the rear Ackermann angle.
13-14	Steering angle, right rear (RR)	int16	raw = value × 1000 (0.001 per bit) rad	±32.767	
15	Ultrasonic collision avoidance	uint8	—	0x00 = Disabled 0x01 = Indoor collision avoidance (30 cm stop distance) 0x02 = Outdoor collision avoidance (50 cm stop distance)	
16	Kinematic model	uint8	—	0x01 = Differential 0x02 = Dual Ackermann 0x03 = Four-wheel steering	Numbering differs from the CAN protocol (CAN: 0x03 four-wheel steering, 0x04 dual Ackermann).
17	Turn indicator	uint8	—	0x00 = No control 0x01 = Left 0x02 = Right 0x03 = Hazard (both)	
18	External serial control	uint8	—	0x00 = External serial control disabled (built-in compute unit) 0x01 = External serial control enabled (external compute unit)	
19	Reserved	uint8	—	0x00	
20	Frame tail	uint8	—	0x1C	Confirm against your firmware release.

Table 5-9 · Feedback frame (Chassis → Host) — 21 bytes

Byte	Field	Type	Scaling / unit	Values	Notes
0	Frame header	uint8	—	0xC2	Confirm against your firmware release.
1-2	Linear velocity x (vx) — resultant speed	int16	raw = value × 1000 (0.001 per bit) m/s	±32.767	
3-4	Linear velocity y (vy)	int16	raw = value × 1000 (0.001 per bit) m/s	±32.767	
5-6	Yaw rate (wz) — resultant yaw rate	int16	raw = value × 1000 (0.001 per bit) rad/s	±32.767	Yaw rate. Earlier documentation lists the unit as m/s; the field is an angular rate in rad/s. Verify on your unit.
7-8	Steering angle, left front (LF)	int16	raw = value × 1000 (0.001 per bit) rad	±32.767	Also used as the front Ackermann angle.
9-10	Steering angle, right front (RF)	int16	raw = value × 1000 (0.001 per bit) rad	±32.767	
11-12	Steering angle, left rear (LR)	int16	raw = value × 1000 (0.001 per bit) rad	±32.767	Also used as the rear Ackermann angle.
13-14	Steering angle, right rear (RR)	int16	raw = value × 1000 (0.001 per bit) rad	±32.767	
15	Battery level	uint8	—	0-100 Low-battery warning below 15	Percent.
16	Kinematic model	uint8	—	0x01 = Differential 0x02 = Dual Ackermann 0x03 = Four-wheel steering	
17	Control mode	uint8	—	0x11 = Remote control 0x22 = Autonomous	
18	Fault status	uint8	—	0x00 = No fault 0x01 = Ultrasonic stop 0x02 = Emergency stop 0x03 = Parking brake engaged 0x04 = Charging (charger connected) 0x05-0x0E = Chassis fault (see sheet "Error Codes")	
19	Reserved	uint8	—	0x00	

Byte	Field	Type	Scaling / unit	Values	Notes
20	Frame tail	uint8	—	0x2C	Confirm against your firmware release.

5.4 ROS software

The ROS software for the Luchs family is supplied as `luchs_ros2` (version 1.1, ROS 2 Humble and Jazzy) and `luchs_ros1` (version 1.1, ROS 1 Noetic). Both are built on the same ROS-independent C++ SDK and drive Luchs A and Luchs B from one code base, with the same topics, frames and services. Publish a velocity on `cmd_vel`; the driver sends the CAN commands, enforces speed, steering and acceleration limits, and stops the chassis when commands or feedback stop.

Table 5-10 · Packages

Package	Content
<code>luchs_sdk</code>	C++ library without ROS: CAN protocols, SocketCAN, steering kinematics, watchdogs, built-in chassis simulators, <code>luchs_cli</code> tool, Python examples
<code>luchs_driver</code>	Driver node; parameter files for A-DF, A-FA, A-DA and A-4WS (and Luchs B)
<code>luchs_msgs</code>	ChassisState, SafetyState, LampCommand, DriveCommand and services
<code>luchs_description</code>	Robot model (URDF) for all four configurations, with mounting frames on the rails
<code>luchs_bringup</code> · <code>luchs_teleop</code>	Launch files, CAN set-up script, start-at-boot units, gamepad configuration

Table 5-11 · Main topics of the driver

Direction	Topic	Type	Content
Subscribed	<code>cmd_vel</code>	<code>geometry_msgs/Twist</code>	Velocity: forward, sideways (A-4WS), yaw rate
Published	<code>odom</code>	<code>nav_msgs/Odometry</code>	Wheel odometry and the <code>odom</code> → <code>base_footprint</code> transform
Published	<code>joint_states</code>	<code>sensor_msgs/JointState</code>	Steering angles and wheel rotation, for the robot model
Published	<code>~/state</code> · <code>~/safety</code>	<code>luchs_msgs/...</code>	Chassis state, faults, lamps; emergency stops, bumpers, ultrasonic ranges
Published	<code>battery_state</code>	<code>sensor_msgs/BatteryState</code>	State of charge, voltage, temperature
Published	<code>/diagnostics</code>	<code>diagnostic_msgs/...</code>	Link status, faults, battery warning

Quick start (ROS 2): bring the CAN interface up at 1 Mbit/s (`setup_can.sh can0`), set the mode selector to automatic, then `ros2 launch lushs_bringup lushs_a.launch.py build:=4ws (or df, fa, da)`. **ROS 1:** `roslaunch lushs_bringup lushs_a.launch build:=4ws`. Without a chassis, add `dry_run:=true` to run against the built-in simulator. The README and the docs folder in each package cover installation, wiring, parameters, sensor integration and troubleshooting.

A few protocol details depend on the chassis firmware release. They are driver parameters (`protocol.*`); check them once per vehicle as described in `docs/protocol_luchs_a.md`.

5.5 Autonomy by TAR (option)

The Luchs family is built for autonomy. TerraAgent Robotics (TAR) supplies it as an option: sensor kit, compute, positioning, perception, planning, vehicle control and the control station, integrated on the Luchs A and delivered as one system. TAR software drives the chassis through the interfaces and the ROS driver described in this chapter, so sections 5.1 to 5.4 apply to a vehicle with the TAR kit as well. Figure 5–1 shows building blocks that already run on real vehicles and real data.

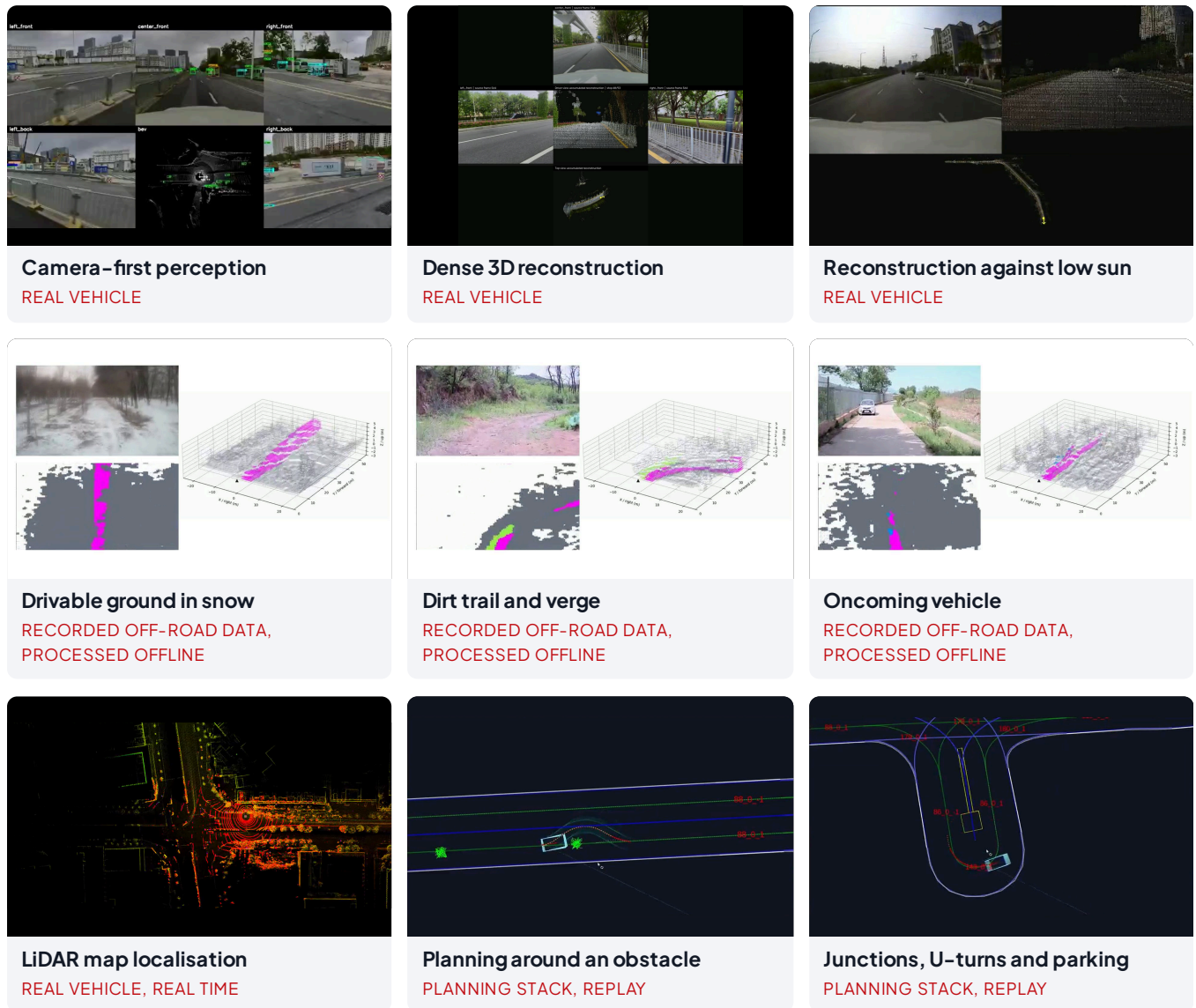


Figure 5–1 · TAR building blocks. Top: camera perception and 3D reconstruction on real vehicles. Middle: off-road drivable ground, our model run offline on recorded off-road driving data. Bottom: LiDAR map localisation on a real vehicle, and replays of the planning stack.

- **Positioning** degrades safely: RTK, then visual-inertial coasting at reduced speed, then a timed safe stop; LiDAR map localisation is the second source when satellites drop out.
- **Perception** is camera-first, with LiDAR and radar added per application: 3D detection, a drivable-ground map and a live elevation map for holes and steps.
- **Planning** runs in three layers (route, behaviour, trajectory) and hands its commands to the chassis through the ROS driver.

The functions, the sensor kit and the delivery for your site are agreed with TAR. Contact details are in section 6.3.



MAINTENANCE AND SUPPORT

6.1 Routine care

- Check the wheels, suspension and fasteners before each use.
- Operate and store the chassis between -10 °C and 45 °C.
- If the chassis is not used for a long time, keep the battery between 50 % and 70 % and charge it regularly.
- Use only original spare parts and the supplied charger.
- If a fault occurs, switch the chassis off and contact support; do not dismantle it yourself.

6.2 Warranty

The warranty terms for your Luchs A are set out in your sales contract with RJ Tech Limited. They exclude normal wear, consumables and damage from misuse, accident or unauthorised modification.

6.3 Support

CHASSIS

RJ Tech Limited

Hardware, drivetrain, firmware, spare parts and warranty.

rjtech-offroad.com

AUTONOMY

TerraAgent Robotics

Sensors, compute, autonomy software and integration.

Contact: Brandon Zhe Feng

When you contact support, please have the chassis configuration (A-DF, A-FA, A-DA or A-4WS), the serial number and a description of the problem ready, including any fault code reported by the chassis (Appendix B).

DIMENSIONS



All dimensions in millimetres.

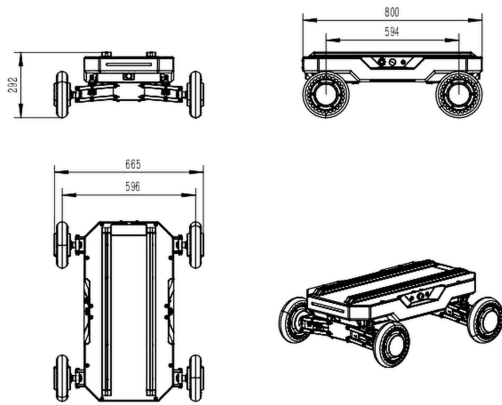


Figure A-1 · Luchs A-DF (differential 4WD)

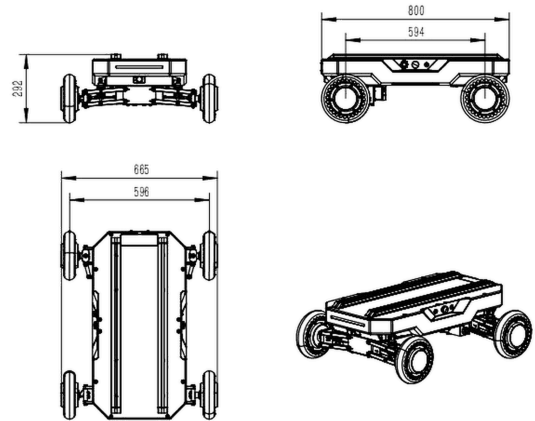


Figure A-2 · Luchs A-FA (front-Ackermann 4WD)

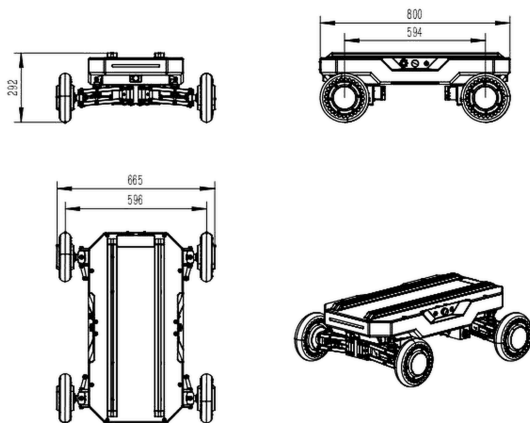


Figure A-3 · Luchs A-DA (dual-Ackermann 4WD)

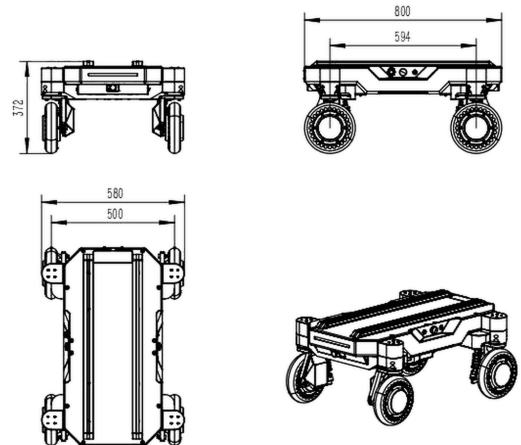


Figure A-4 · Luchs A-4WS (four-wheel steering, standard build)



FAULT CODES

Chassis fault codes are reported in CAN message 0x5B8 (byte 3) and in the fault-status byte of every serial feedback frame. Codes 0x00 to 0x04 appear in the serial protocol only.

Code	Meaning
0x00	No fault
0x01	Ultrasonic stop triggered
0x02	Emergency stop triggered
0x03	Parking brake engaged
0x04	Charging (charger connected; motion disabled)
0x05	Left-front wheel drive fault (left drive on single-motor-per-side builds)
0x06	Right-front wheel drive fault (right drive on single-motor-per-side builds)
0x07	Left-rear wheel drive fault
0x08	Right-rear wheel drive fault
0x09	Left-front steering motor fault (front steering motor on single-steering-motor builds)
0x0A	Right-front steering motor fault
0x0B	Left-rear steering motor fault
0x0C	Right-rear steering motor fault
0x0D	Remote-control timeout in remote-control mode (transmitter out of range or switched off)
0x0E	Watchdog triggered



TERMS AND ABBREVIATIONS

4WD	Four-wheel drive: every wheel has its own drive motor.
4WS	Four-wheel steering: every wheel can be steered on its own.
Ackermann steering	Car-style steering in which the inner wheel turns more sharply than the outer one, so both follow the same arc.
BMS	Battery Management System: electronics that monitor and protect the battery.
CAN	Controller Area Network: a robust two-wire bus widely used in vehicles and machines.
DB9	9-pin D-subminiature connector.
Host computer	The computer that commands the chassis, such as the built-in computer or your industrial PC.
IMU	Inertial Measurement Unit: measures acceleration and rotation rate.
IP33	Ingress Protection rating to IEC 60529: protected against solid objects over 2.5 mm and against spraying water.
LiFePO₄	Lithium iron phosphate: a lithium-ion battery chemistry known for long cycle life and thermal stability.
MCU	Microcontroller unit: the main controller of the chassis.
Odometry	An estimate of the robot's motion calculated from wheel speeds and steering angles.
Pivot turn	Turning on the spot about the centre of the chassis.
ROS	Robot Operating System: widely used open-source robotics middleware.
RS-232 / RS-485	Serial communication standards. RS-485 is differential and suits longer, noisier cable runs.
SDK	Software Development Kit.
SLAM	Simultaneous Localisation and Mapping: building a map while working out where the robot is on it.
SocketCAN	The standard CAN interface of the Linux kernel.